

Physics 513 QFT HW 7

Pavel Okun

Department of Chemistry, University of Michigan, Ann Arbor USA

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CONTENTS

1. 1
2. 2
3. 3

- 1
- 1
- 4

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int \frac{d^3k_2 d^3k_3 d^3k_4}{k_2^0 k_3^0 k_4^0} \delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) (\vec{k}_2 \cdot \vec{k}_4). \quad (7)$$

$$E_i = k_i^0 \quad (8)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int \frac{d^3k_2 d^3k_3 d^3k_4}{E_2 E_4} \delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) (\vec{k}_2 \cdot \vec{k}_4) \quad (9)$$

1. 1

I looked over the notes.

2. 2

$$\delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) = \delta(E_1 - E_2 - E_3 - E_4) \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \quad (10)$$

$$d\Gamma = \delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) \frac{d^3k_2 d^3k_3 d^3k_4 |\mathcal{M}|^2}{2^4 m (2\pi)^5 k_2^0 k_3^0 k_4^0} \quad (1)$$

$$|\mathcal{M}|^2 = 64g^2 (\vec{k}_1 \cdot \vec{k}_3) (\vec{k}_2 \cdot \vec{k}_4) \quad (2)$$

$$d\Gamma = \frac{4g^2}{m(2\pi)^5} \quad (11)$$

$$\delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) \frac{d^3k_2 d^3k_3 d^3k_4}{k_2^0 k_3^0 k_4^0} (\vec{k}_1 \cdot \vec{k}_3) (\vec{k}_2 \cdot \vec{k}_4) \quad (12)$$

$$k_i = |\mathbf{k}_i|, \quad E_2 = k_2, \quad E_4 = k_4 \quad (12)$$

The masses are

$$m_1 = m, \quad m_2 = m_3 = m_4 = 0. \quad (13)$$

(3)

In the decaying particle's rest frame

$$\vec{k}_1 = (m, 0, 0, 0). \quad (4)$$

To get the total decay rate I integrate over the final (decay product) 3-momenta

$$\Gamma = \frac{4g^2}{m(2\pi)^5} \int \frac{d^3k_2 d^3k_3 d^3k_4}{k_2^0 k_3^0 k_4^0} \delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) (\vec{k}_1 \cdot \vec{k}_3) (\vec{k}_2 \cdot \vec{k}_4). \quad (5)$$

The energies k_i^0 are not independent of the 3-momenta: they are functions of 3-momentum and must be included in the integral.

$$\vec{k}_1 \cdot \vec{k}_3 = (m, \mathbf{0}) \cdot (k_3^0, \mathbf{k}_3) = m k_3^0 \quad (6)$$

In SR the energy is a function of the 3-momentum.

$$\delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) = \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \delta \left(\sqrt{k_1^2 + m^2} - \sqrt{k_2^2 + m^2} - \sqrt{k_3^2 + m^2} - \sqrt{k_4^2 + m^2} \right)$$

$$\delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) = \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \delta \left(\sqrt{k_1^2 + m^2} - \sqrt{k_2^2} - \sqrt{k_3^2} - \sqrt{k_4^2} \right) \quad (14)$$

$$\delta^{(4)}(\vec{k}_1 - \vec{k}_2 - \vec{k}_3 - \vec{k}_4) = \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \delta \left(\sqrt{k_1^2 + m^2} - k_2 - k_3 - k_4 \right) \quad (15)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3k_2 d^3k_3 d^3k_4 \frac{(\vec{k}_2 \cdot \vec{k}_4)}{k_2 k_4} \delta \left(\sqrt{k_1^2 + m^2} - k_2 - k_3 - k_4 \right) \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \quad (16)$$

$$\vec{k}_2 \cdot \vec{k}_4 = (E_2, \mathbf{k}_2) \cdot (E_4, \mathbf{k}_4) = E_2 E_4 - \mathbf{k}_2 \cdot \mathbf{k}_4 = k_2 k_4 - \mathbf{k}_2 \cdot \mathbf{k}_4 \quad (17)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3 k_2 d^3 k_3 d^3 k_4 \left(1 - \frac{\mathbf{k}_2 \cdot \mathbf{k}_4}{k_2 k_4} \right) \delta \left(\sqrt{k_1^2 + m^2} - k_2 - k_3 - k_4 \right) \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \quad (18)$$

We are working in the frame where $\mathbf{k}_1 = \mathbf{0}$ so

$$\mathbf{k}_1 = \mathbf{0} \implies k_1 = 0. \quad (19)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3 k_2 d^3 k_3 d^3 k_4 \left(1 - \frac{\mathbf{k}_2 \cdot \mathbf{k}_4}{k_2 k_4} \right) \delta(m - k_2 - k_3 - k_4) \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4) \quad (20)$$

$$\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_3 - \mathbf{k}_4 = 0 \implies \mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_4 = \mathbf{k}_3 \implies k_3 = |\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_4| = |\mathbf{0} - \mathbf{k}_2 - \mathbf{k}_4| = |\mathbf{k}_2 + \mathbf{k}_4| \quad (21)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3 k_2 d^3 k_4 \left(1 - \frac{\mathbf{k}_2 \cdot \mathbf{k}_4}{k_2 k_4} \right) \delta(m - k_2 - |\mathbf{k}_2 + \mathbf{k}_4| - k_4) \quad (22)$$

I change the integration variables to get rid of the subscripts.

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3 k d^3 p \left(1 - \frac{\mathbf{k} \cdot \mathbf{p}}{kp} \right) \delta(m - k - |\mathbf{k} + \mathbf{p}| - p) \quad (23)$$

$$I(\mathbf{v}) = \int d^3 p \left(1 - \frac{\mathbf{v} \cdot \mathbf{p}}{vp} \right) \delta(m - v - |\mathbf{v} + \mathbf{p}| - p) \quad (24)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3 k I(\mathbf{k}) \quad (25)$$

To integrate $I(\mathbf{v})$ I choose coordinates where $\mathbf{v} = v \hat{\mathbf{z}}$.

$$I(\mathbf{v}) = \int d^3 p \left(1 - \frac{\hat{\mathbf{z}} \cdot \mathbf{p}}{p} \right) \delta(m - v - |\mathbf{v} + \mathbf{p}| - p) \quad (26)$$

$$I(\mathbf{v}) = \int d^3 p \left(1 - \frac{p_z}{p} \right) \delta(m - v - |\mathbf{v} + \mathbf{p}| - p) \quad (27)$$

$$\mathbf{v} + \mathbf{p} = (0, 0, v) + (p_x, p_y, p_z) = (p_x, p_y, p_z + v) \quad (28)$$

$$|\mathbf{v} + \mathbf{p}| = \sqrt{p_x^2 + p_y^2 + (p_z + v)^2} = \sqrt{p_x^2 + p_y^2 + p_z^2 + v^2 + 2vp_z} = \sqrt{p^2 + v^2 + 2vp_z} \quad (29)$$

$$I(v) = \int d^3 p \left(1 - \frac{p_z}{p} \right) \delta \left(m - v - \sqrt{p^2 + v^2 + 2vp_z} - p \right) \quad (30)$$

I relabel the dummy variables $\mathbf{p} = (p_x, p_y, p_z) \rightarrow \mathbf{r} = (x, y, z)$.

$$I(v) = \int d^3 x \left(1 - \frac{z}{r} \right) \delta \left(m - v - \sqrt{r^2 + v^2 + 2vz} - r \right) \quad (31)$$

I use spherical coordinates so $z = r \cos \theta$.

$$I(v) = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \int_0^\infty dr r^2 (1 - \cos \theta) \delta \left(m - v - \sqrt{r^2 + v^2 + 2vr \cos \theta} - r \right) \quad (32)$$

$$I(v) = 2\pi \int_0^\pi d\theta \sin \theta \int_0^\infty dr r^2 (1 - \cos \theta) \delta \left(m - v - \sqrt{r^2 + v^2 + 2vr \cos \theta} - r \right) \quad (33)$$

$$u = \cos \theta, \quad u(0) = \cos 0 = 1, \quad u(\pi) = \cos \pi = -1, \quad \frac{du}{d\theta} = -\sin \theta \implies -du = d\theta \sin \theta \quad (34)$$

$$I(v) = 2\pi \int_{-1}^1 du \int_0^\infty dr r^2 (1 - u) \delta \left(m - v - \sqrt{r^2 + v^2 + 2vru} - r \right) \quad (35)$$

$$\begin{aligned} m - v - \sqrt{r^2 + v^2 + 2vru} - r &= 0 \implies \\ m - v - r &= \sqrt{r^2 + v^2 + 2vru} \implies \\ r^2 + v^2 + 2vru &= (m - v - r)^2 = \\ m^2 + r^2 + v^2 - 2mr - 2mv + 2rv &\implies \\ u &= m \frac{m - 2r - 2v}{2rv} + 1 \end{aligned} \quad (36)$$

I also have $-1 \leq u \leq 1$ from Eq. 35. So when I use the delta function to eliminate u I will need to put in the appropriate Heaviside theta functions to enforce this assumption. Because r and v represent vector magnitudes they are positive.

$$\begin{aligned} -1 \leq u &\implies 4rv \geq m(2r+2v-m) \\ \boxed{u \leq 1 \implies m \leq 2(r+v)} \end{aligned} \quad (37)$$

The second constraint is a nontrivial condition that must be enforced with a theta function. The first constraint can be rewritten

$$\begin{aligned} 4rv \geq m(2r+2v-m) &\implies -4rv \leq m^2 - 2m(r+v) \implies \\ &-4rv = 2(r-v)^2 - 2r^2 - 2v^2 \leq \\ m^2 - 2m(r+v) &= (m-r-v)^2 - r^2 - v^2 - 2rv \implies \\ 2(r-v)^2 - r^2 - v^2 + 2rv &\leq (m-r-v)^2 \implies \\ 2(r-v)^2 - (r-v)^2 &\leq (m-r-v)^2 \implies (r-v)^2 \leq (m-r-v)^2 \\ |r-v| &\leq |m-r-v|. \end{aligned} \quad (38)$$

Though I have relabeled variables several times it should be clear that r and v are 3-momenta magnitudes of 2 massless decay products. That means r and v are also energies. And conservation of energy implies

$$m > r + v, \quad (39)$$

because r and v are only 2 of the 3 products m decayed into. Together Eqs. 38 & 39 imply

$$|r-v| \leq m-r-v \implies r+v+|r-v| = 2\max(r,v) \leq m, \quad (40)$$

implies

$$\boxed{m \geq 2r}, \quad \boxed{m \geq 2v}. \quad (41)$$

Now I evaluate the delta function in Eq. 35, making sure to put in the three conditions in Eqs. 37 & 41 as theta functions. All of these conditions are implicit in the delta function, which encodes conservation of energy, so no new information is introduced. And one more thing: I also need to rewrite the delta function in a new form using the identity:

$$\delta[f(x)] = \frac{\delta(x-x_0)}{|f'(x_0)|}, \quad (42)$$

where $f(x)$ has one root, x_0 , so $f(x_0) = 0$. There is another generalization for functions with more than one root, but this is all I need here.

$$f(u) = m - v - \sqrt{r^2 + v^2 + 2rvu} - r \quad (43)$$

$$u_0 = m \frac{m-2r-2v}{2rv} + 1 \quad (44)$$

$$\frac{1}{|f'(u)|} = \frac{\sqrt{r^2 + v^2 + 2rvu}}{rv} \quad (45)$$

Since r and v are 3-vector magnitudes they are positive.

$$\begin{aligned} \delta\left(m - v - \sqrt{r^2 + v^2 + 2rvu} - r\right) &= \Theta(m-2v) \\ \Theta[2(r+v)-m]\Theta(m-2r)\delta(u-u_0) &\frac{\sqrt{r^2 + v^2 + 2rvu_0}}{rv} \end{aligned} \quad (46)$$

$$\begin{aligned} I(v) &= \frac{2}{v}\pi\Theta(m-2v) \int_{-1}^1 du \int_0^\infty dr r(1-u)\delta(u-u_0) \\ &\sqrt{r^2 + v^2 + 2rvu_0}\Theta[2(r+v)-m]\Theta(m-2r) \end{aligned} \quad (47)$$

$$\begin{aligned} I(v) &= \frac{2}{v}\pi\Theta(m-2v) \int_0^\infty dr r(1-u_0)\sqrt{r^2 + v^2 + 2rvu_0} \\ &\Theta[2(r+v)-m]\Theta(m-2r) \end{aligned} \quad (48)$$

$$\begin{aligned} I(v) &= \frac{\pi}{v^2}m\Theta(m-2v) \int_{m/2-v}^{m/2} dr (2r+2v-m) \\ &\sqrt{r^2 + v^2 + 2rvu_0} \end{aligned} \quad (49)$$

$$\begin{aligned} I(v) &= \frac{\pi}{v^2}m\Theta(m-2v) \int_{m/2-v}^{m/2} dr (2r+2v-m) \\ &\sqrt{(r+v-2m)(r+v)+m^2} \end{aligned} \quad (50)$$

$$I(v) = \pi m \Theta(m-2v) \left(\frac{m}{2} - \frac{2}{3}v \right) \quad (51)$$

$$\Gamma = \frac{4g^2}{(2\pi)^5} \int d^3k I(k) = \frac{8g^2}{(2\pi)^4} \int_0^\infty dk k^2 I(k) \quad (52)$$

$$\Gamma = \frac{4g^2 m}{(2\pi)^3} \int_0^\infty dk k^2 \Theta(m-2k) \left(\frac{m}{2} - \frac{2}{3}k \right) \quad (53)$$

$$\Gamma = \frac{4g^2 m}{(2\pi)^3} \int_0^{m/2} dk k^2 \left(\frac{m}{2} - \frac{2}{3}k \right) \quad (54)$$

$$\boxed{\Gamma = \frac{g^2 m^5}{192\pi^3}} \quad (55)$$

3. 3

$$|\mathcal{M}|^2 = e^4 \left(\frac{\vec{p} \cdot \vec{k}}{\vec{p} \cdot \vec{k}'} + \frac{\vec{p} \cdot \vec{k}'}{\vec{p} \cdot \vec{k}} + 2m^2 A + m^4 A^2 \right) \quad (56)$$

$$A = \frac{1}{\vec{p} \cdot \vec{k}} - \frac{1}{\vec{p} \cdot \vec{k}'} \quad (57)$$

$$\vec{p} \cdot \vec{p} = \vec{p}' \cdot \vec{p}' = m^2, \quad \vec{k}^2 = \vec{k}'^2 = 0 \quad (58)$$

$$k^0 = \omega, \quad k'^0 = \omega' \quad (59)$$

4-momentum conservation yields

$$\vec{p} + \vec{k} = \vec{p}' + \vec{k}'. \quad (60)$$

Squaring both sides yields

$$(\vec{p} + \vec{k})^2 = (\vec{p}' + \vec{k}')^2, \quad (61)$$

$$\vec{p}^2 + \vec{k}^2 + 2\vec{p} \cdot \vec{k} = \vec{p}'^2 + \vec{k}'^2 + 2\vec{p}' \cdot \vec{k}', \quad (62)$$

$$\cancel{m^2} + 2\vec{p} \cdot \vec{k} = \cancel{m^2} + 2\vec{p}' \cdot \vec{k}', \quad (63)$$

$$\vec{p} \cdot \vec{k} = \vec{p}' \cdot \vec{k}'. \quad (64)$$

Rearranging Eq. 60 and squaring yields

$$(\vec{p}' - \vec{p})^2 = (\vec{k} - \vec{k}')^2, \quad (65)$$

$$\vec{p}'^2 + \vec{p}^2 - 2\vec{p}' \cdot \vec{p} = \vec{k}^2 + \vec{k}'^2 - 2\vec{k} \cdot \vec{k}', \quad (66)$$

$$m^2 + \vec{k} \cdot \vec{k}' = \vec{p}' \cdot \vec{p}. \quad (67)$$

Rearranging and squaring Eq. 60 also yields

$$(\vec{p} - \vec{k}')^2 = (\vec{p}' - \vec{k})^2, \quad (68)$$

$$\vec{p}^2 + \vec{k}'^2 - 2\vec{p} \cdot \vec{k}' = \vec{p}'^2 + \vec{k}^2 - 2\vec{p}' \cdot \vec{k}, \quad (69)$$

$$\cancel{m^2} - 2\vec{p} \cdot \vec{k}' = \cancel{m^2} - 2\vec{p}' \cdot \vec{k}, \quad (70)$$

$$\vec{p} \cdot \vec{k}' = \vec{p}' \cdot \vec{k}. \quad (71)$$

$$(\vec{p} + \vec{k}) \cdot \vec{k} = (\vec{p}' + \vec{k}') \cdot \vec{k} \quad (72)$$

$$\vec{p} \cdot \vec{k} + \vec{k}^2 = \vec{p}' \cdot \vec{k} + \vec{k}' \cdot \vec{k} \quad (73)$$

$$\vec{p} \cdot \vec{k} = \vec{p}' \cdot \vec{k} + \vec{k}' \cdot \vec{k} \quad (74)$$

I use Eq. 71.

$$\vec{p} \cdot \vec{k}' = \vec{p} \cdot \vec{k} - \vec{k}' \cdot \vec{k} \quad (75)$$

The right frame for this problem is the massless particle $\vec{k}$ collides with the $\vec{p}$ (which is at rest) and then a massive ($\vec{p}'$) and massless ($\vec{k}'$) particle leave. The momenta are

$$\begin{aligned} \vec{p} &= (m, 0, 0, 0), \\ \vec{k} &= \omega(1, 1, 0, 0), \\ \vec{p}' &= (E, p'_x, p'_y, 0), \\ \vec{k}' &= \omega'(1, \cos \theta, \sin \theta, 0), \end{aligned} \quad (76)$$

where θ is the angle with the direction of motion of $\vec{k}$ (the x axis).

$$A = \frac{1}{\vec{p} \cdot \vec{k}} - \frac{1}{\vec{p} \cdot \vec{k}'} \quad (77)$$

$$A = \frac{1}{m\omega} - \frac{1}{m\omega'} \quad (78)$$

$$mA = \frac{1}{\omega} - \frac{1}{\omega'} \quad (79)$$

Conservation of y momentum implies

$$p'_y = -\omega' \sin \theta. \quad (80)$$

Conservation of x momentum implies

$$\omega = p'_x + \omega' \cos \theta, \quad (81)$$

$$p'_x = \omega - \omega' \cos \theta. \quad (82)$$

$$\begin{aligned} \vec{p} &= (m, 0, 0, 0), \\ \vec{k} &= \omega(1, 1, 0, 0), \\ \vec{p}' &= (E, \omega - \omega' \cos \theta, -\omega' \sin \theta, 0), \\ \vec{k}' &= \omega'(1, \cos \theta, \sin \theta, 0), \end{aligned} \quad (83)$$

$$E = \sqrt{m^2 + \mathbf{p}^2} \quad (84)$$

$$E = \sqrt{m^2 + p_x'^2 + p_y'^2} \quad (85)$$

$$E = \sqrt{m^2 + (\omega - \omega' \cos \theta)^2 + \omega'^2 \sin^2 \theta} \quad (86)$$

$$E = \sqrt{m^2 + \omega^2 + \omega'^2 \cos^2 \theta - 2\omega\omega' \cos \theta + \omega'^2 \sin^2 \theta} \quad (87)$$

$$E = \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta} \quad (88)$$

I apply conservation of energy.

$$m + \omega = E + \omega' \quad (89)$$

$$E = m + \omega - \omega' \quad (90)$$

$$E^2 = (m + \omega - \omega')^2 \quad (91)$$

$$\begin{aligned} \cancel{m^2 + \omega^2 + \omega'^2} - 2\omega\omega' \cos \theta = \\ \cancel{m^2 + \omega^2 + \omega'^2} + 2(m\omega - m\omega' - \omega\omega') \end{aligned} \quad (92)$$

$$(m + \omega - \omega \cos \theta)\omega' = m\omega \quad (93)$$

$$\boxed{\omega' = \frac{m\omega}{m + \omega(1 - \cos \theta)}} \quad (94)$$

This expression is identical to Eq. (5) in the problem set. This confirms that I set up the frame (Eq. 76) correctly.

$$mA = \frac{1}{\omega} - \frac{1}{\omega'} \quad (95)$$

$$mA = \frac{1}{\omega} - \frac{m + \omega(1 - \cos \theta)}{m\omega} \quad (96)$$

$$m^2 A = \cos \theta - 1 \quad (97)$$

$$|\mathcal{M}|^2 = e^4 \left(\frac{\vec{p} \cdot \vec{k}}{\vec{p} \cdot \vec{k}'} + \frac{\vec{p} \cdot \vec{k}'}{\vec{p} \cdot \vec{k}} + 2m^2 A + m^4 A^2 \right) \quad (98)$$

$$|\mathcal{M}|^2 = e^4 \left[\frac{\vec{p} \cdot \vec{k}}{\vec{p} \cdot \vec{k}'} + \frac{\vec{p} \cdot \vec{k}'}{\vec{p} \cdot \vec{k}} + 2 \cos \theta - 2 + (\cos \theta - 1)^2 \right] \quad (99)$$

$$\boxed{\frac{|\mathcal{M}|^2}{e^4} = \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2 \theta} \quad (100)$$

Equation (5.22) in Schwartz defines the infinitesimal cross section for a collision between two particles. Applying that equation to this problem yields

$$d\sigma = \frac{|\mathcal{M}|^2 d\Pi_{\text{LIPS}}}{4m\omega v_{\text{rel}}} = \frac{|\mathcal{M}|^2 d\Pi_{\text{LIPS}}}{4m\omega} \quad (101)$$

where v_{rel} is the relative velocity between the 2 initial particles. In this case $v_{\text{rel}} = 1$, the speed of light, and

$$d\Pi_{\text{LIPS}} = \frac{d^3 p' d^3 k'}{16\pi^2 E \omega'} \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}' - \vec{k}'). \quad (102)$$

$$d\sigma = d^3 p' d^3 k' \frac{|\mathcal{M}|^2}{64\pi^2 m E \omega'} \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}' - \vec{k}') \quad (103)$$

$$64\pi^2 m\omega \int d\sigma = \int d^3 p' d^3 k' \frac{|\mathcal{M}|^2}{E \omega'} \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}' - \vec{k}') \quad (104)$$

$$64\pi^2 m\omega\sigma = \int d^3 p' d^3 k' \frac{|\mathcal{M}|^2}{E \omega'} \delta^{(4)}(\vec{p} + \vec{k} - \vec{p}' - \vec{k}') \quad (105)$$

I relabel the momenta $\vec{p}' \rightarrow \vec{q}$ and $\vec{k}' \rightarrow \vec{v}$, but I have to remember that the energy E is now associated with $\vec{q}$ and ω' with $\vec{v}$.

$$64\pi^2 m\omega\sigma = \int d^3 q d^3 v \frac{|\mathcal{M}|^2}{E \omega'} \delta^{(4)}(\vec{p} + \vec{k} - \vec{q} - \vec{v}) \quad (106)$$

$$\begin{aligned} 64\pi^2 m\omega\sigma = \int d^3 q d^3 v \frac{|\mathcal{M}|^2}{E \omega'} \delta^{(3)}(\mathbf{p} + \mathbf{k} - \mathbf{q} - \mathbf{v}) \\ \delta(m + \omega - E - \omega') \end{aligned} \quad (107)$$

$$E = \sqrt{m^2 + \mathbf{q}^2} = \sqrt{m^2 + q^2} \quad (108)$$

$$\begin{aligned} 64\pi^2 m\omega\sigma = \int d^3 q d^3 v \frac{|\mathcal{M}|^2}{\omega' \sqrt{m^2 + q^2}} \delta^{(3)}(\mathbf{p} + \mathbf{k} - \mathbf{q} - \mathbf{v}) \\ \delta\left(m + \omega - \omega' - \sqrt{m^2 + q^2}\right) \end{aligned} \quad (109)$$

$$\mathbf{p} = \mathbf{0}, \quad \mathbf{k} = \omega \hat{\mathbf{x}} \quad (110)$$

$$\begin{aligned} 64\pi^2 m\omega\sigma = \int d^3 q d^3 v \frac{|\mathcal{M}|^2}{\omega' \sqrt{m^2 + q^2}} \delta^{(3)}(\mathbf{k} - \mathbf{q} - \mathbf{v}) \\ \delta\left(m + \omega - \omega' - \sqrt{m^2 + q^2}\right) \end{aligned} \quad (111)$$

$$\mathbf{k} - \mathbf{q} - \mathbf{v} = \mathbf{0} \implies \mathbf{k} - \mathbf{v} = \mathbf{q} \quad (112)$$

$$q^2 = \mathbf{q}^2 = (\mathbf{k} - \mathbf{v})^2 = \mathbf{k}^2 + \mathbf{v}^2 - 2\mathbf{k} \cdot \mathbf{v} \quad (113)$$

$$\mathbf{v} = \omega'(\cos \theta, \sin \theta, 0) \quad (114)$$

$$\mathbf{k}^2 = \omega^2 \quad \mathbf{v}^2 = \omega'^2 \quad (115)$$

$$q^2 = \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta \quad (116)$$

$$64\pi^2 m\omega\sigma = \int d^3v \frac{|\mathcal{M}|^2}{\omega' \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta}} \delta \left(m + \omega - \omega' - \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta} \right) \quad (117)$$

$$\mathbf{v}^2 = \omega'^2 \quad \& \quad 0 < \omega' \implies v = \omega' \quad (118)$$

$$64\pi^2 m\omega\sigma = \int d^3v \frac{|\mathcal{M}(v)|^2}{v \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta}} \delta \left(m + \omega - v - \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta} \right) \quad (119)$$

Since this is a radial function of v I can bring in the solid angle now by switching to spherical coordinates.

$$64\pi^2 m\omega\sigma = \int d\Omega dv v^2 \frac{|\mathcal{M}(v)|^2}{v \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta}} \delta \left(m + \omega - v - \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta} \right) \quad (120)$$

$$64\pi^2 m\omega\sigma = \int d\Omega dv \frac{v |\mathcal{M}(v)|^2}{\sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta}} \delta \left(m + \omega - v - \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta} \right) \quad (121)$$

$$f(v) = m + \omega - v - \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta} \quad (122)$$

$$64\pi^2 m\omega\sigma = \int_0^{4\pi} d\Omega \int_0^\infty dv \frac{v |\mathcal{M}(v)|^2 \delta[f(v)]}{\sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta}} \quad (123)$$

$$v_0 = \omega' = \frac{m\omega}{m + \omega(1 - \cos \theta)}, \quad f(v_0) = 0 \quad (124)$$

Note that I have to be very careful to not create any ambiguity between $v = \omega'$ the integration variable and $v_0 = \omega'$ the specific value. This is why I introduced the v and v_0 symbols. From now on ω' is always the specific value.

$$\delta[f(v)] = \frac{\delta(v - v_0)}{|f'(v_0)|} = \frac{\delta(v - \omega')}{|f'(\omega')|} \quad (125)$$

$$64\pi^2 m\omega\sigma = \int_0^{4\pi} d\Omega \int_0^\infty dv \frac{v |\mathcal{M}(v)|^2 \delta(v - \omega')}{|f'(\omega')| \sqrt{m^2 + \omega^2 + v^2 - 2\omega v \cos \theta}} \quad (126)$$

$$64\pi^2 m\omega\sigma = \int_0^{4\pi} d\Omega \frac{\omega' |\mathcal{M}(\omega')|^2}{|f'(\omega')| \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta}} \quad (127)$$

$$64\pi^2 m\omega d\sigma = \frac{d\Omega \omega' |\mathcal{M}|^2}{|f'(\omega')| \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta}} \quad (128)$$

Remembering that $m + \omega > \omega'$ (conservation of energy) I show (after much algebra which I outsourced to Mathematica) that

$$\frac{1}{|f'(\omega')|} = \frac{m + \omega - \omega'}{m\omega} \omega', \quad \sqrt{m^2 + \omega^2 + \omega'^2 - 2\omega\omega' \cos \theta} = m + \omega - \omega'. \quad (129)$$

$$\frac{d\sigma}{d\Omega} = \left(\frac{\omega'}{\omega} \right)^2 \frac{|\mathcal{M}|^2}{64\pi^2 m^2} \quad (130)$$

$$\boxed{\frac{d\sigma}{d\Omega} = \left(\frac{e^2 \omega'}{8\pi m\omega} \right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2 \theta \right)} \quad (131)$$

This does not match Eq. (4) in the problem set but it does match the solution. So there is probably a typo in Eq. (4). QED.